SAYAK DUTTA

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PROFILE

Senior AI/ML Engineer with 10+ years of experience, I build AI that ships and stays up. Specialised in multi-agent orchestration (LangGraph, LangChain), hybrid RAG architectures, and real-time distributed systems on AWS and GCP. Rooted in data engineering, entity resolution, NLP pipelines, high-volume data systems, which gives me the infrastructure rigour most AI engineers skip. I work at the intersection of AI architecture and product delivery, partnering with product managers and business stakeholders to turn capability into outcomes: 60% latency reduction, 45% LLM cost savings, 99.9% production reliability across live systems.

PROFESSIONAL EXPERIENCE

Siaris GmbH, *AI Engineer* 2025 – Present | Germany

- Architected and shipped the company's core multi-agent LLM platform (LangGraph + LangChain) from scratch, now the primary production system powering workflows for 100+ daily active users, live within 3 months of joining.
- Redesigned agent orchestration on AWS Lambda + ECS; cut median response latency from 12s to under 6s, directly driving a 38% increase in weekly active usage.
- Designed hybrid RAG with HNSW vector retrieval + semantic re-ranking; LangSmith-based evaluation harness measured a 42% improvement in context relevance post-rollout.
- Implemented hallucination auditing, permission-scoped retrieval, and PII-safe data isolation; system running at 99%+ reliability and scaling toward 99.9%.
- Cut LLM inference costs by 45% through chunking optimisation, Redis retrieval caching, and intelligent model routing, zero quality degradation.
- Sole technical voice in roadmap discussions with founding team and enterprise clients; architecture decisions made directly from business requirements.

Torto Labs Pvt Ltd., *Senior Machine Learning Engineer* 2021 – 2025 | India (Remote)

- Led delivery of an enterprise RAG assistant on AWS (S3 + Lambda + RDS) combining FAISS vector retrieval, structured PostgreSQL access, and LangChain + OpenAI API, processing 1M+ records/month for natural language analytics.
- Slashed analyst lookup latency from 60s to under 20s (ElastiCache + async inference); grew product adoption from 22% to 61% within 6 months.
- Built the team's model evaluation framework in LangSmith, tracking retrieval recall, grounding accuracy, and latency SLAs; became the mandatory release gate for all AI features.
- Architected event-driven inference pipelines on AWS MSK (Kafka) + ElastiCache (Redis), 99.9% uptime under variable production load.
- Established CI/CD via GitHub Actions + AWS CodeDeploy; cut deployment cycle time by 60%+ and moved the team to weekly release cadence.
- Partnered with product managers and client teams to define AI feature scope; mentored 2 junior engineers on RAG patterns and LLM evaluation.

Berylls GmbH, *Machine Learning Engineer* 2023 – 2023 | Germany

- Productionised collaborative filtering models into FastAPI endpoints on GCP Cloud Run, 100K+ prediction requests/month at p95 latency under 80ms.
- Improved Feature engineering and model optimisation delivering a 25% engagement uplift and 15% conversion improvement on recommendation-driven journeys, validated via A/B testing.
- Built automated drift monitoring + retraining pipelines on GCP Monitoring, eliminated manual model health checks entirely.
- Containerised the full ML stack with Docker + GCP Artifact Registry; reduced MLOps overhead by 80% with fully reproducible deployments.

Nailbiter Inc., *Senior Data Analyst* 2020 – 2021 | India

- Built NLP pipelines for semantic analysis of consumer survey and social data, extracting sentiment signals and theme clusters from unstructured text at scale.
- Automated text classification and entity tagging, replacing manual workflows; 20% targeting accuracy improvement and 30% faster data processing.
- Built a multi-source insights aggregation pipeline cutting campaign strategy turnaround from days to hours for client teams

eClerx Services Ltd., *Analyst* 2018 – 2020 | India

- Designed entity resolution and fuzzy matching systems (Levenshtein + phonetic) processing 500K+ records/month, 35% accuracy improvement across client product catalogues.
- Automated compliance reporting pipelines, 40% reduction in manual effort, turnaround cut from 3 days to same-day.

EDUCATION

Bachelor of Engineering– Mechanical Engineering, *University of Mumbai* 2012 – 2016 | Mumbai, India
Grade: First Division

SKILLS

AI / LLM
LLMs (OpenAI, Anthropic, OSS), RAG & Hybrid Retrieval, Multi-Agent Systems, LangChain, LangGraph, LangSmith, FAISS, pgvector, Qdrant, Embeddings, Prompt Engineering, Model Evaluation

ML / Data
PyTorch, Hugging Face Transformers, Scikit-learn, NLP, Recommendation Systems, Time-Series, Feature Engineering

Full-Stack
Python, TypeScript / JavaScript, FastAPI, Node.js, Next.js, REST APIs, WebSockets, PostgreSQL

Cloud
AWS (Lambda, ECS, S3, RDS, MSK, ElastiCache, CodeDeploy), GCP (Cloud Run, Artifact Registry, Monitoring)

Infrastructure
Docker, Kubernetes, Kafka, Redis, Celery, GitHub Actions CI/CD, Serverless, MLOps, Model Monitoring

PROJECTS

ResearchRAG - Academic PDF Research Assistant
- Architected a RAG pipeline with custom chunking strategy (sliding window + section-aware splitting) handling complex academic PDF structure, figures, citations, methodology sections, more accurately than naive approaches.
- FAISS dense retrieval + BM25 hybrid re-ranking via LangChain; LangSmith evaluation tracing. Delivers structured outputs in under 30 seconds on 30-page papers, 3-5× faster than manual workflows.
- Tech stack: FastAPI · Next.js · TypeScript · FAISS · BM25 · LangChain · LangSmith · PyMuPDF · OpenRouter OSS

AskPostgres - Natural Language to SQL for Business Intelligence
- Two-stage pipeline: LLM generates SQL → validation layer checks syntax, schema binding, and injection risk before execution, no hallucinated queries reach production.
- Schema-level RBAC enforced at query generation time; cut analyst SQL-writing time by up to 80%.
- Tech stack: FastAPI · Next.js · TypeScript · PostgreSQL · pgvector · LangChain · OpenRouter

OutreachPilot - Async Campaign Automation & Analytics Platform
- Event-driven orchestration with Celery + Redis queues, decoupled send scheduling from delivery tracking; dead-letter queues + retry policies for fault tolerance.
- Custom pixel + redirect tracking middleware for open rates, CTR, and A/B variant performance at individual recipient level.
- Tech stack: FastAPI · Next.js · TypeScript · PostgreSQL · Celery · Redis · Docker · Resend API

TalkToTube - Long-Form Video Knowledge Retrieval
- Segments transcripts by semantic topic boundaries (not fixed time windows), coherent chunks from hour-long videos.
- LangChain QA chain over FAISS vector store; answers cite source timestamps so users jump directly to the relevant segment.
- Tech stack: FastAPI · Next.js · TypeScript · PostgreSQL · FAISS · LangChain · Docker · YouTube Transcript API

LANGUAGES AND VISA

Languages:
English (Fluent), Hindi (Native), German (A2)

Visa:
Valid in Germany; Open to relocation

Work Authorization:
EU Work Authorization; Open to employer-sponsored visas